

Notation Reference

Notation Reference I

This paper intentionally minimizes mathematical notation to maximize accessibility. Where notation is used, it follows the Cognitive Integrity Framework (CIF) formal specification defined in Part 1 of this series.

Minimal Notation Used I

Symbol	Meaning	Plain Language
n	Trust decay factor	"Delegated trust decreases by this factor at each step"
f	Agent count	"Number of agents in the system"
$[A]$	Byzantine agents	"Maximum number of malicious agents tolerated"
$\{FR\}$	Design Matrix	Maps Functional Requirements to Defense Provisions
$\{DP\}$	Functional Requirements	What the system must protect
Ω_k	Defense Provisions	What CIF mechanisms provide
	Adversary class k	Capability tier (1=passive, 2=injection, 3=spoofing, 4=belief manipulation, 5=coordinated)

CIF-AD-OODA Notation I

The cross-domain analysis (Sections 9c–9l) uses the CIF-AD-OODA methodology:

- ▶ **Design Matrix** $[A]$: A matrix where rows represent Functional Requirements (FR) and columns represent Defense Provisions (DP). Each entry A_{ij} indicates whether defense j covers requirement i .
- ▶ **Transient Coupling** $[A']$: The coupling matrix during an active attack, showing which defenses are bypassed.
- ▶ **Adversary Classes** Ω_1 – Ω_5 : Five capability tiers from passive eavesdropping (Ω_1) through coordinated multi-agent attacks (Ω_5), as defined in Part 1, Section 3.

CIF-AD-OODA Notation II

Three Universal Attack Patterns

Across all ten domains, attacks reduce to three canonical patterns:

Pattern	Description	Defense
FR Polarity Inversion	Attacker flips a Functional Requirement's sign (e.g., "don't share secrets" → "share secrets")	Cognitive Firewall + Belief Sandbox
Constraint Relaxation	Attacker weakens a safety constraint's boundary	Invariant Monitor + Tripwire
Context Boundary Violation	Attacker exploits scope leakage between agent contexts	Provenance Tracking + Trust Calculus

Trust Decay Explanation I

When we write $\gamma = 0.9$, this means:

- ▶ Direct trust: 100% of assigned value
- ▶ One delegation: 90% of source trust
- ▶ Two delegations: 81% of source trust
- ▶ Three delegations: 73% of source trust

A lower γ (e.g., 0.85) means faster decay, providing more security but limiting delegation utility.

Byzantine Tolerance Explanation I

When we say $n \geq 3f + 1$:

- ▶ To tolerate 1 malicious agent, need at least 4 agents
- ▶ To tolerate 2 malicious agents, need at least 7 agents
- ▶ To tolerate 3 malicious agents, need at least 10 agents

Full Notation Reference I

For complete formal definitions of all CIF notation, see Part 1 supplementary **S03** (`cogsec_multiagent_1_theory/manuscript/S03_notation.md` in this repository's `cognitive_integrity` program tree).

The Part 1 specification uses on the order of 100 symbols covering:

- ▶ Agent cognitive state
- ▶ Trust calculus operations
- ▶ Defense mechanism parameters
- ▶ Consensus and coordination
- ▶ Information-theoretic bounds